SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Solution Architecture

Date	5 July 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

The Solution Architecture illustrates the overall structure of the OptiCrop Smart Agricultural Production Optimization System. It consists of the Presentation Layer for user interaction, the Application Layer for processing data and generating crop predictions using machine learning models, and the Data Layer for storing user information, datasets, and prediction results.

Presentation / Client Layer

HTML
CSS
JavaScript
Web Browser

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API Gateway / Load Balancer

Flask (Python)

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Auth Service

User Login & Registration

Core Logic Service

Data Preprocessing
Machine Learning Model (KNN, Logistic Regression, Decision Tree, Random Forest)
Crop Prediction

External API Integrations

Kaggle Dataset
Weather API *(Optional)*

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Data / Storage Layer

MySQL Database
Crop Dataset (.csv)
Trained ML Model (.pkl)

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly interface to enter soil details and display crop recommendations.	HTML, CSS, JavaScript
API Gateway	Receives requests from users and forwards them to the backend services.	Flask
Microservices	Handles authentication, data preprocessing, machine learning prediction, and report generation.	Python, Scikit-learn, Pandas, NumPy
Database	Stores user information, soil data, crop details, and prediction results.	MySQL